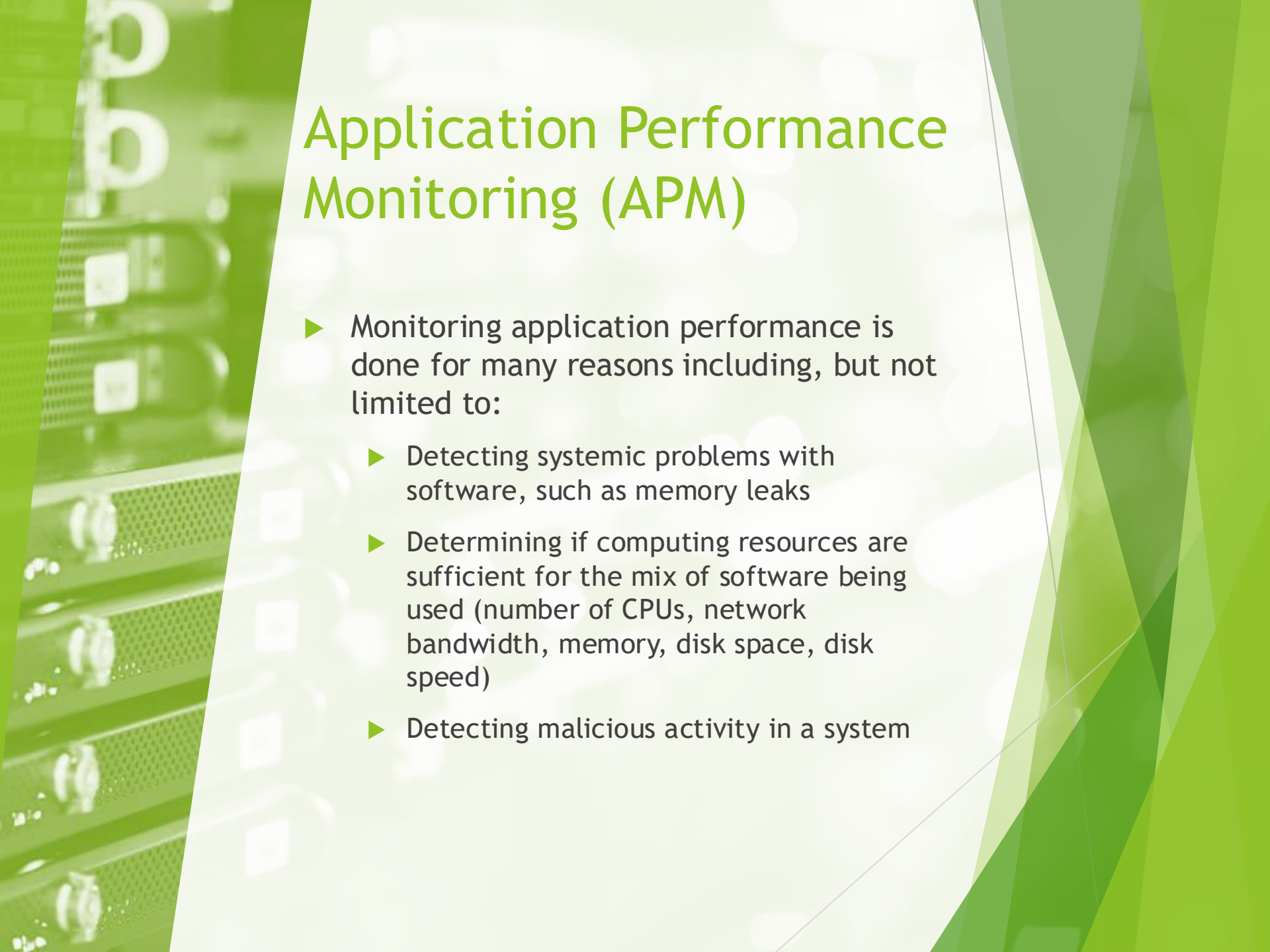




NSSA-220 Project 1

Application Performance Monitoring



Application Performance Monitoring (APM)

- ▶ Monitoring application performance is done for many reasons including, but not limited to:
 - ▶ Detecting systemic problems with software, such as memory leaks
 - ▶ Determining if computing resources are sufficient for the mix of software being used (number of CPUs, network bandwidth, memory, disk space, disk speed)
 - ▶ Detecting malicious activity in a system

Points of Inspiration

APM is highly relevant to just about any career path in computing, especially systems engineering, network management, network and system security, and system administration

Employers are impressed by anyone that has experience with APM and analyzing APM data is highly related to the field of data analytics/data science

Project 1 Preliminaries

Done in teams of 3

Declare your team on
myCourses discussion area

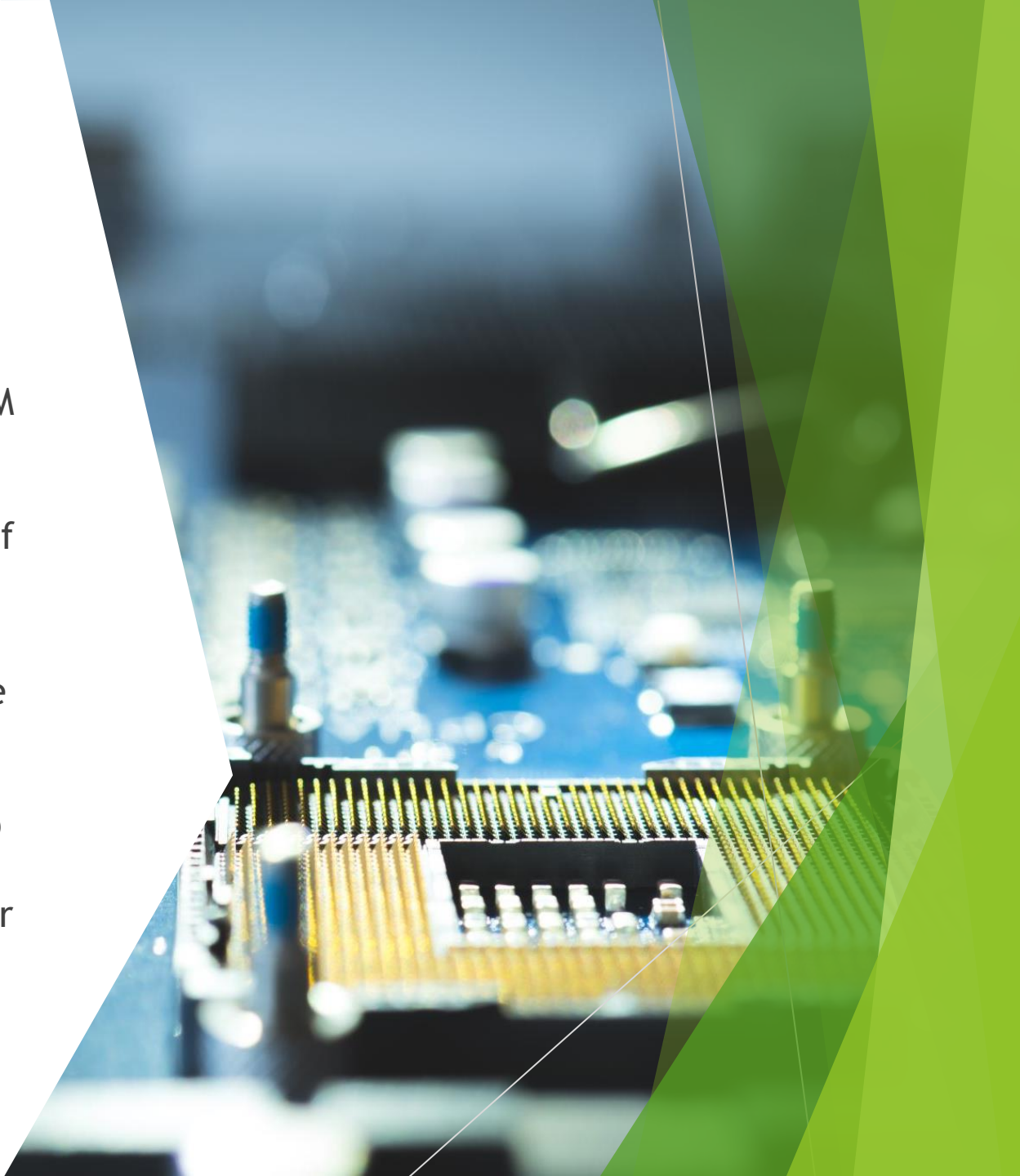
Yes, you must do this in teams!
No lone wolves allowed!



Projects are designed to be more difficult than labs and require your own research beyond what you've been taught so far (worth 15% of your grade)

APM Tool Overview

- ▶ You'll be writing an APM tool in Bash that will monitor a mix of processes in the form of executable programs written in C
- ▶ The script will start the processes, collect performance metrics, and perform a clean up at the end to kill these processes and any other processes that your script spawns



APM Tool Overview (continued)



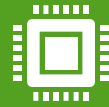
The APM tool will monitor application performance by collecting metrics at both:

Process level
System level



Your final submission will include a short report with Excel plots of the performance metrics collected

Process Level Metrics



Linux provides the ability to collect performance metrics at the process level for both CPU and memory



Using ps, you can see the %CPU and %memory that any process is using

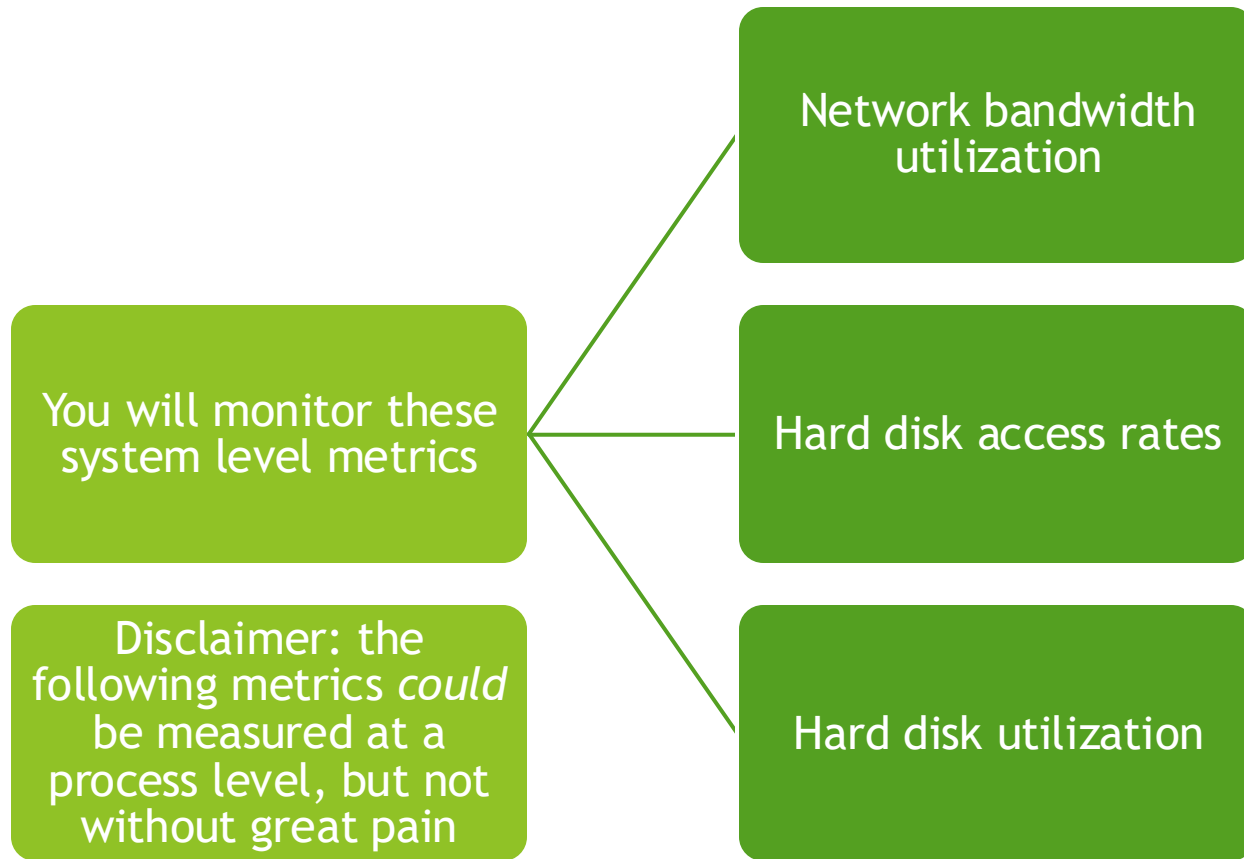


If the applications are utilizing most of or all of the CPU, you may consider adding cores to a VM or processors to a physical machine



Unbounded memory growth over time indicates poor memory management

System Level Metrics



Network Bandwidth Utilization



Network bandwidth utilization indicates how much bandwidth each network interface card is using over time



You will measure bandwidth utilization in terms of receive (RX) data rate and transmit (TX) data rate using the ifstat tool



If data rates are approaching network capacity, additional capacity and/or load balancing is likely needed



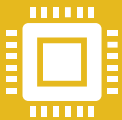
Hard Disk Access Rates

- ▶ Hard disk access rates are typically measured in volume of reads and writes per second
- ▶ Monitoring disk access activity over time may show that a solid state drive is needed over a mechanical drive or alert a system administrator to unexpected excessive access rates
- ▶ You will measure hard disk writes in kB/second to the primary hard drive (sda) using the iostat tool

Hard Disk Utilization



Monitoring hard disk utilization, in terms of remaining disk capacity, is not as critical as it once was, but there are still times where unexpected *reductions* or *increases* in disk capacity may occur



You will measure hard disk utilization on the “/” mount (the `/dev/mapper/centos-root` filesystem) using the `df` tool

CentOS VM Resources

The default CentOS VM
resources are

1 CPU core

2 GB
memory

100 GB
hard disk

CentOS
NIC
(ens33) in
NAT mode



You are required to use the
default CentOS VM resources
to collect data for this project
MAYBE...

Applications to Monitor



You will be given 6 applications in the form of pre-compiled C executables



Each application will take a single argument: the IP address of a VMNet NIC on the lab PCs or any virtual NIC on your own computer



You will need to run your VM in “NAT Mode” to allow connectivity between the VM and the host’s virtual NIC. The virtual NIC is initially disabled.

APM Tool Requirements (1)

1. The APM tool shall collect process and system level metrics every 5 seconds
2. The APM tool shall collect %CPU and %memory utilization *per process* using the ps tool
3. The APM tool shall collect network bandwidth utilization in terms of RX data rate and TX data rate (kB/s) with a *sampling interval of 1 second* on the ens33 interface using the ifstat tool

More APM Tool Requirements (2)

4. The APM tool shall collect hard disk writes in kB/second to the primary hard drive (sda) using the iostat tool
5. The APM tool shall collect hard disk utilization of the “/” mount in *Megabytes available* using the df tool
6. The APM tool shall output all CPU and memory metrics to a CSV file specific to the process they were measured from. Name the files <proc_name>_metrics.csv

More APM Tool Requirements (3)

7. The format of the process specific output files shall be
 - ▶ <seconds>, <%CPU>, <%memory>
8. The APM tool shall write all system level metrics to a file called `system_metrics.csv`
9. The format of the system level output file shall be
 - ▶ <seconds>, <RX data rate>, <TX data rate>, <disk writes>, <available disk capacity>

More APM Tool Requirements (4)

- 10. The APM tool shall spawn all application processes
- 11. The APM tool shall kill all application processes and any other processes it spawns in an exit trap function called “cleanup”
- 12. The APM tool shall minimally include functions to (1) spawn applications and other processes, (2) collect process level metrics, and (3) collect system level metrics

More APM Tool Requirements (5)

13. Your team's submission to myCourses shall be a single zip file including the APM tool script, all output files for a **15 minute run**, and a report showing Excel plots of the metrics collected. The report format is provided.
14. Your application performance data shall be measured using the default VMWare resources

Grading

- ▶ See grading sheet for details
- ▶ Major point loss will occur if an individual exhibits poor teamwork. You will fill out a peer review form to indicate how well the team worked together.
- ▶ Let me know ASAP if there are issues with your team.



Hints



Read the man pages for the specified tools!



One of the tools requires you to spawn an additional process to use it.



All the values in the output files should be numbers only. You already know what the units are.



Put all the requirements into a table in a Word document and check them off as you complete them.

Ask for help!

1

Don't suffer in silence. Ask me or your TA for help sooner rather than later!

- Attend my office hours or the TA's
- Make an appointment outside of office hours
- Send an email

2

If you're not sure if you've met a specific requirement, please ask!